Project Update Report

Overview

In the Logic Puzzle Solver project, this report describes my individual progress. My main tasks included setting up the development environment, organizing Sudoku test data, developing basic input and output methods, and getting ready for future testing.

My work provides the support needed to load, show, and test Sudoku puzzles effectively, even if the primary solving algorithms are still under development.

Week 1 (24th June – 30th June): Establishing the Development Environment

- **Repository Setup:** I assisted with setting up the shared repository and arranging folders for source code, datasets, tests, documentation, and results.
- **Development Tool Selection:** My team and I selected Python as the primary programming language due to its simplicity and suitability for algorithm development and testing.
- **Coding Standards:** I helped to develop basic coding standards such as clear variable names, structured functions, relevant comments and consistent file names.
- **Project Documentation:** I helped with the project proposal, project scope, and expected features of the Sudoku solver.

The work provided a well defined and ordered environment for the team.

Week 2 (1st July – 7th July): Preparing Sudoku Test Data

- **Puzzle Collection**: I have collected Sudoku puzzles from easy, medium, hard levels
- **Data Organization:** The puzzles have been organized by difficulty level to allow the team to easily select them during testing.
- **Storage Format:** I designed simple text and CSV formats where each puzzle is represented as a 9 by 9 grid, with zeros for blank cells.

- **Data Consistency**: I checked the puzzles for consistency to make sure they were in the correct grid format and could be read correctly by the program.
- **Testing Preparation:** I prepared the dataset to be applied later for testing solution accuracy, solving time and solver performance**.**

This provided a reusable dataset for development and assessment

Week 3 (8th July – 14th July): Building Input and Display Support

- **Puzzle Input:** I developed the fundamental framework for loading Sudoku puzzles from structure text files, CSV files, and strings.
- **Input Processing:** I wrote logic to turn the puzzle data into a structured 9 x 9 grid.
- **Console Display:** I developed a basic Console Display that shows clearly rows, columns, and 3 × 3 sections.
- **Output Verification:** I used the display function to check that the puzzle values were loaded in the right positions.
- **Future Integration:** I created the input and display modules so that they might be connected to the backtracking and constraint propagation algorithms in the future.

In the following process, I will enhance the input validation, write the automated test cases, and test the final solver using the prepared sudoku dataset.